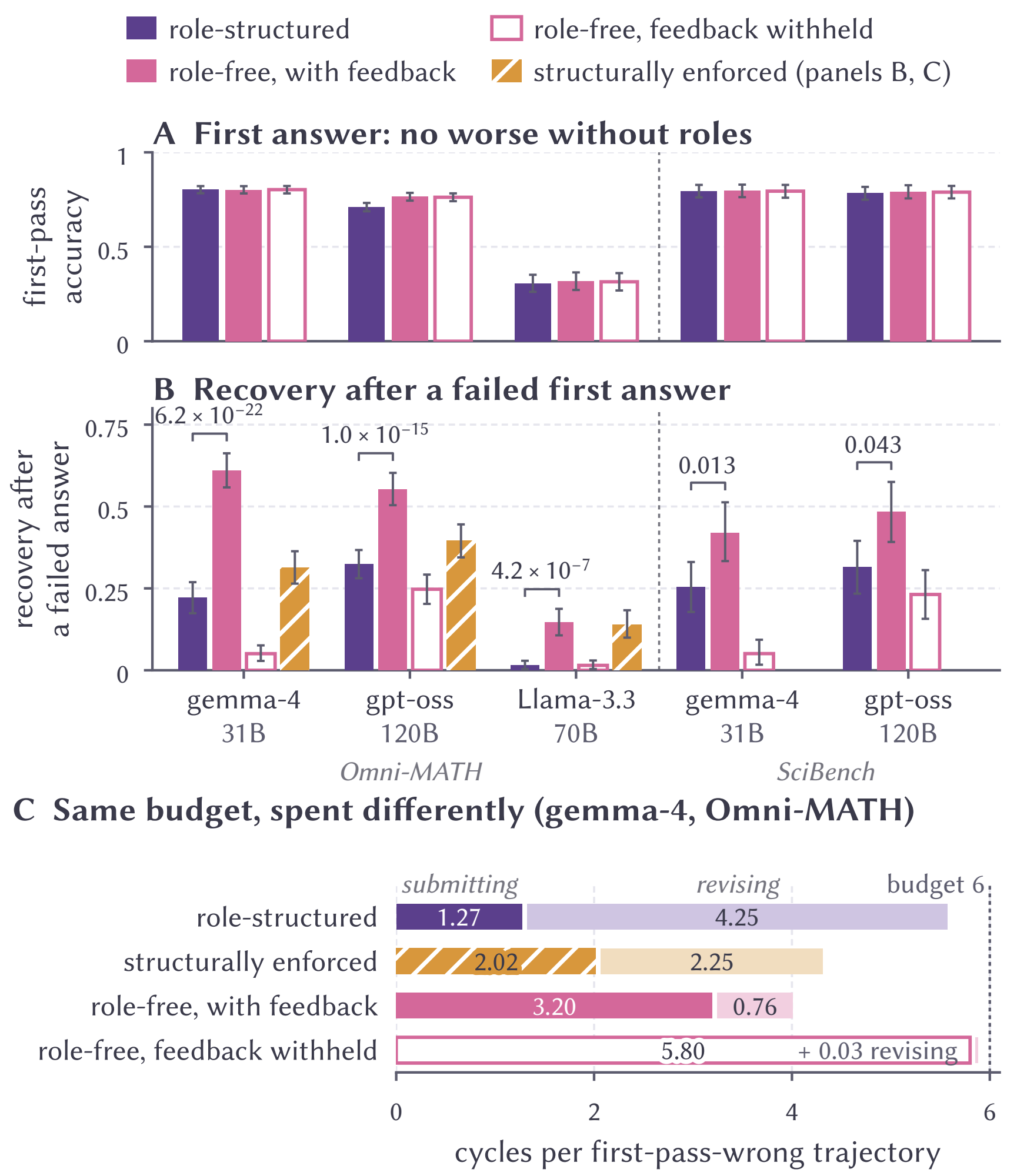
role-structured
role-free, with feedback
role-free, feedback withheld
structurally enforced (panels B, C)
A First answer: no worse without roles
first-pass accuracy
1
0.5
0
B Recovery after a failed first answer
recovery after a failed answer
0.75
0.5
0.25
0
6.2 × 10−22
1.0 × 10−15
4.2 × 10−7
0.013
0.043
gemma-4 31B
gpt-oss 120B
Llama-3.3 70B
gemma-4 31B
gpt-oss 120B
Omni-MATH
SciBench
C Same budget, spent differently (gemma-4, Omni-MATH)
submitting
revising
budget 6
role-structured
1.27
4.25
structurally enforced
2.02
2.25
role-free, with feedback
3.20
0.76
role-free, feedback withheld
5.80
+ 0.03 revising
0
2
4
6
cycles per first-pass-wrong trajectory